



Reduction of the worst-case execution time of real-time tasks targeting many-core platforms

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A need of computing power

Introduction

Data flow graph

Transformations

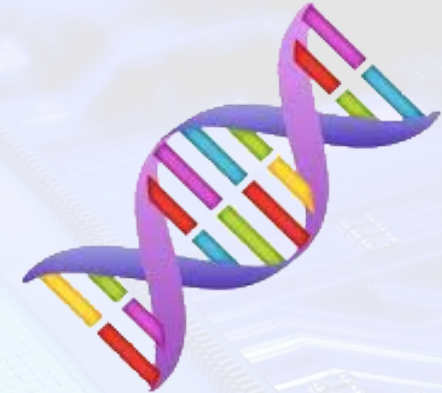
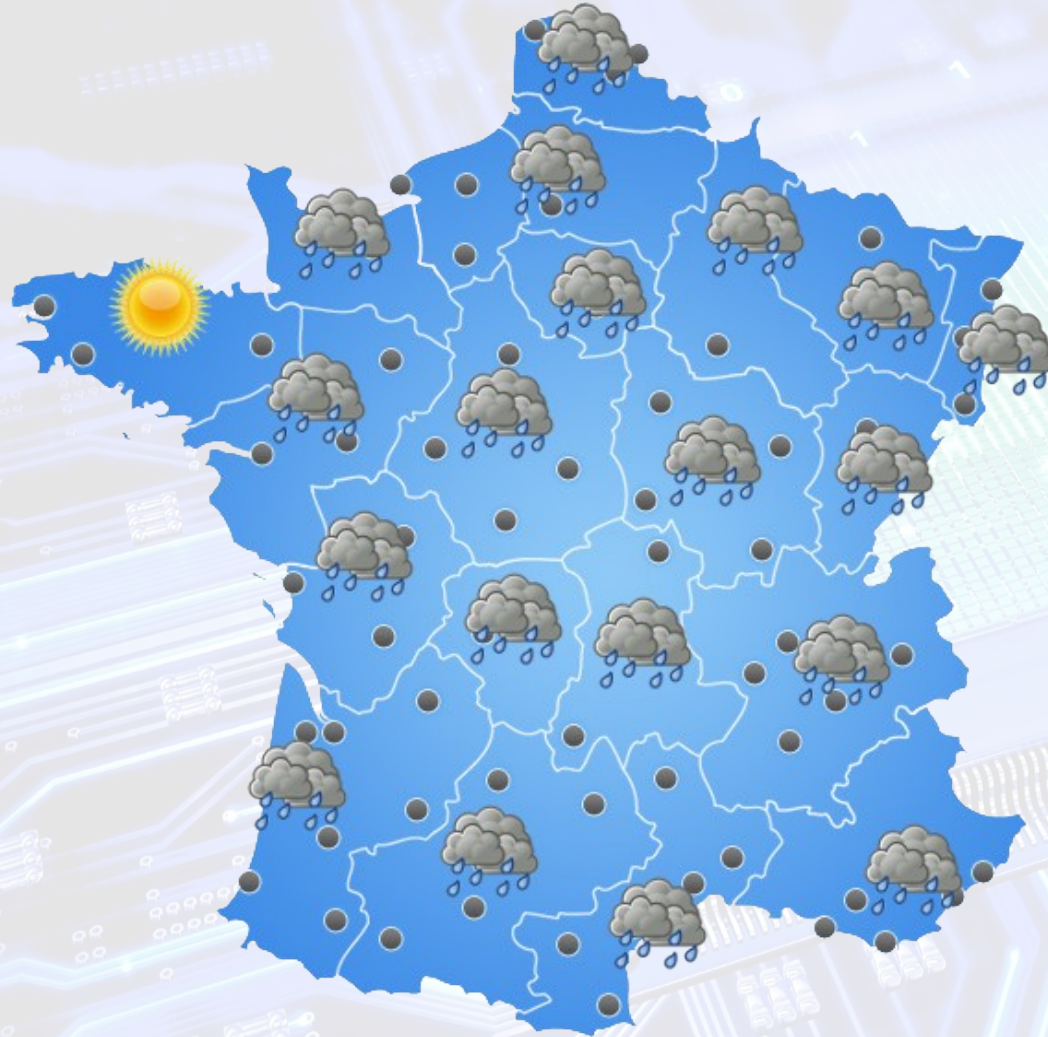
Conclusion

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The end of miniaturization

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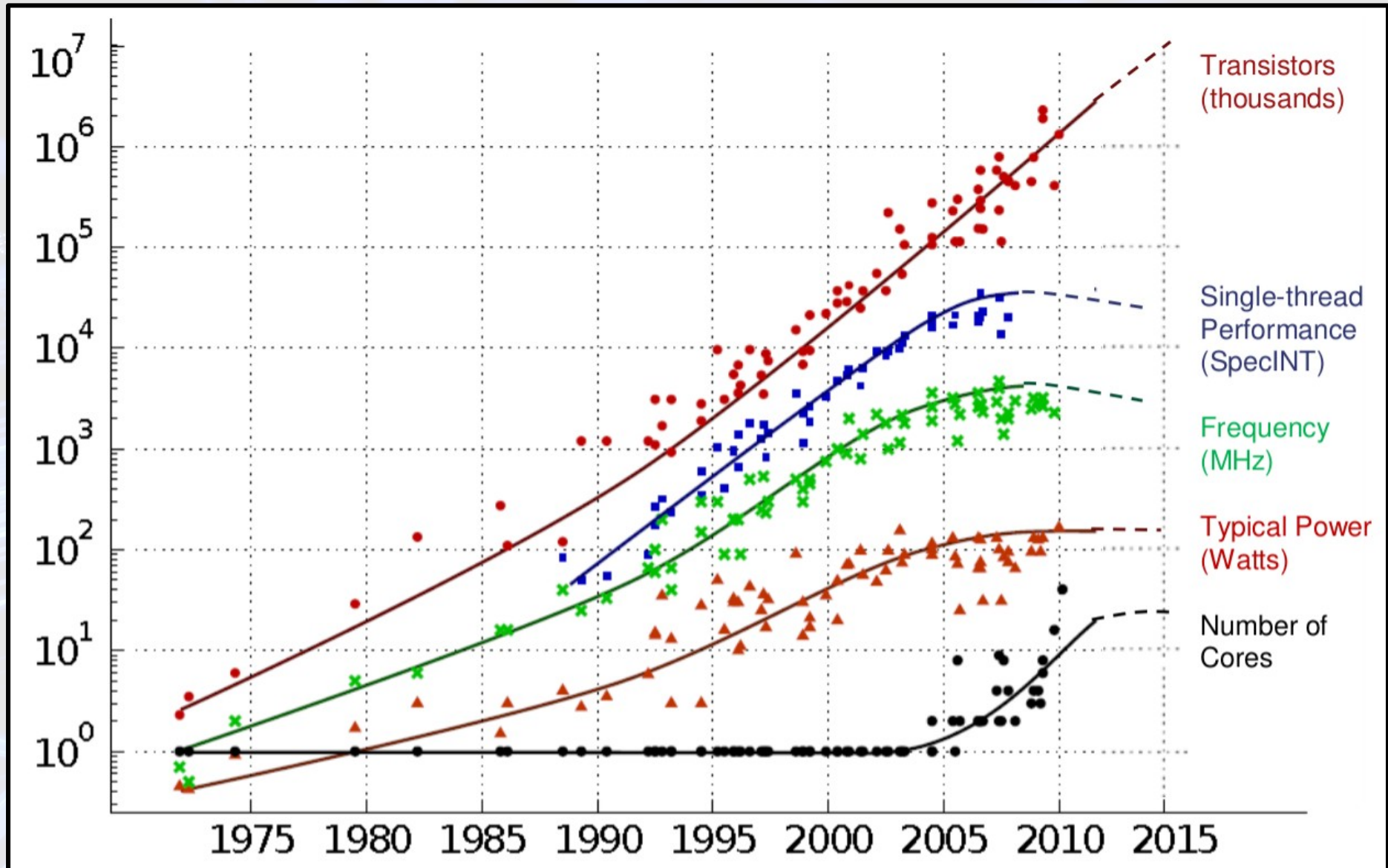
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Current trends

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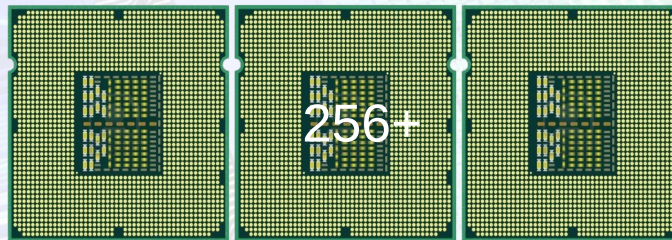
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Real-time programs

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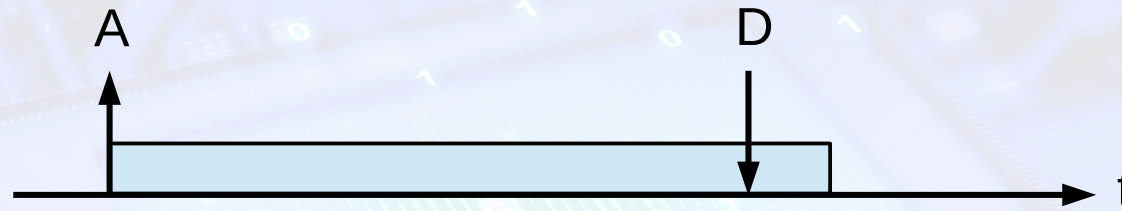
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Reduction of the WCET

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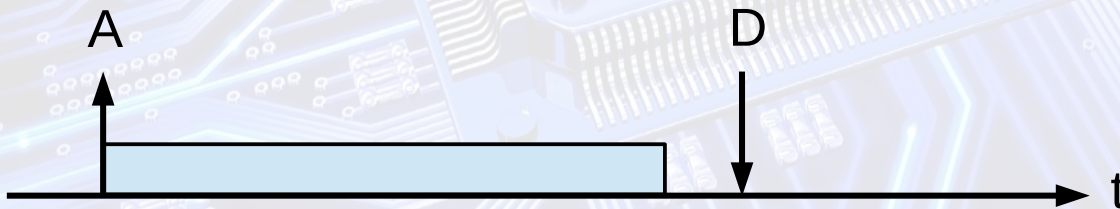
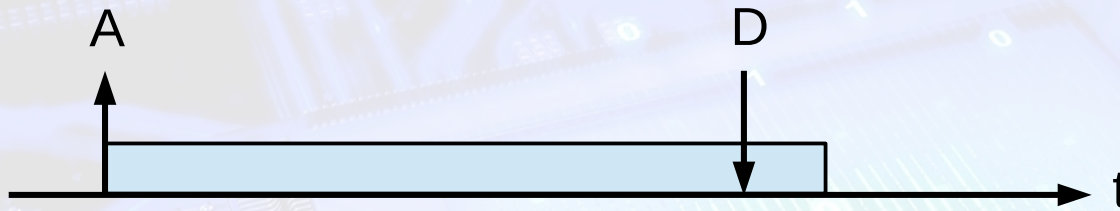
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Some strong assumptions

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- $\text{var}_d := \text{var}_{s1} [\bullet \text{var}_{s2}]$
- Every instruction executes in one unit of time
- No loops or recursion
- Infinite number of cores
- No overheads

Organization of the talk

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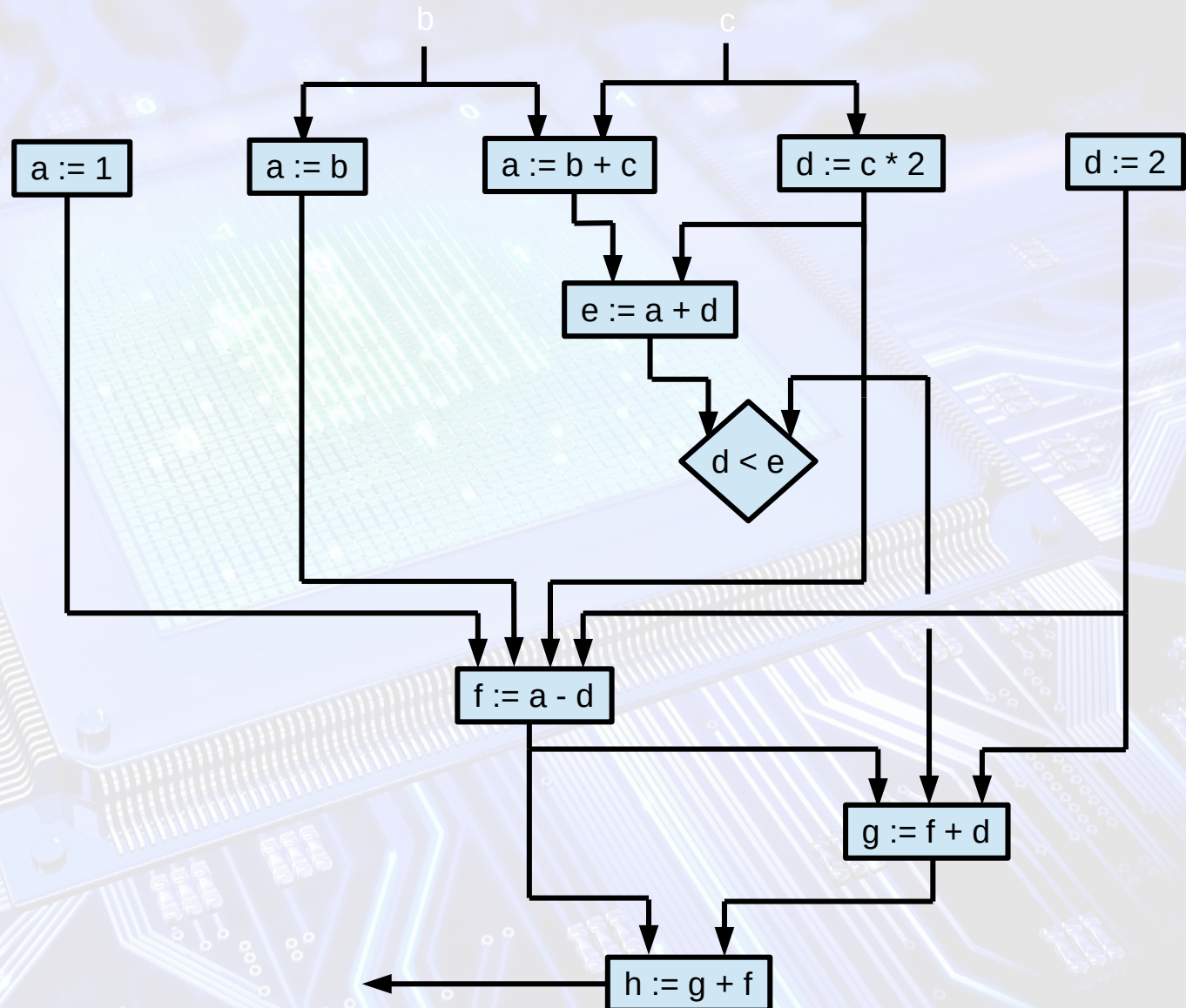
- Data flow graph
 - Data dependencies graph
 - The ambiguous entries
 - Insertion of the selectors
- Transformations
 - Time loss analysis
 - Reduction of the time loss

Precedence constraints between instructions

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Data dependencies graph
The ambiguous entries
Insertion of the selectors

```
function (b, c)
{
  a := b + c
  d := c * 2
  e := a + d
  if (d < e)
    a := 1
  else
    a := b
    d := 2
  f := a - d
  g := f + d
  h := g + f
}
```



Not the same semantics

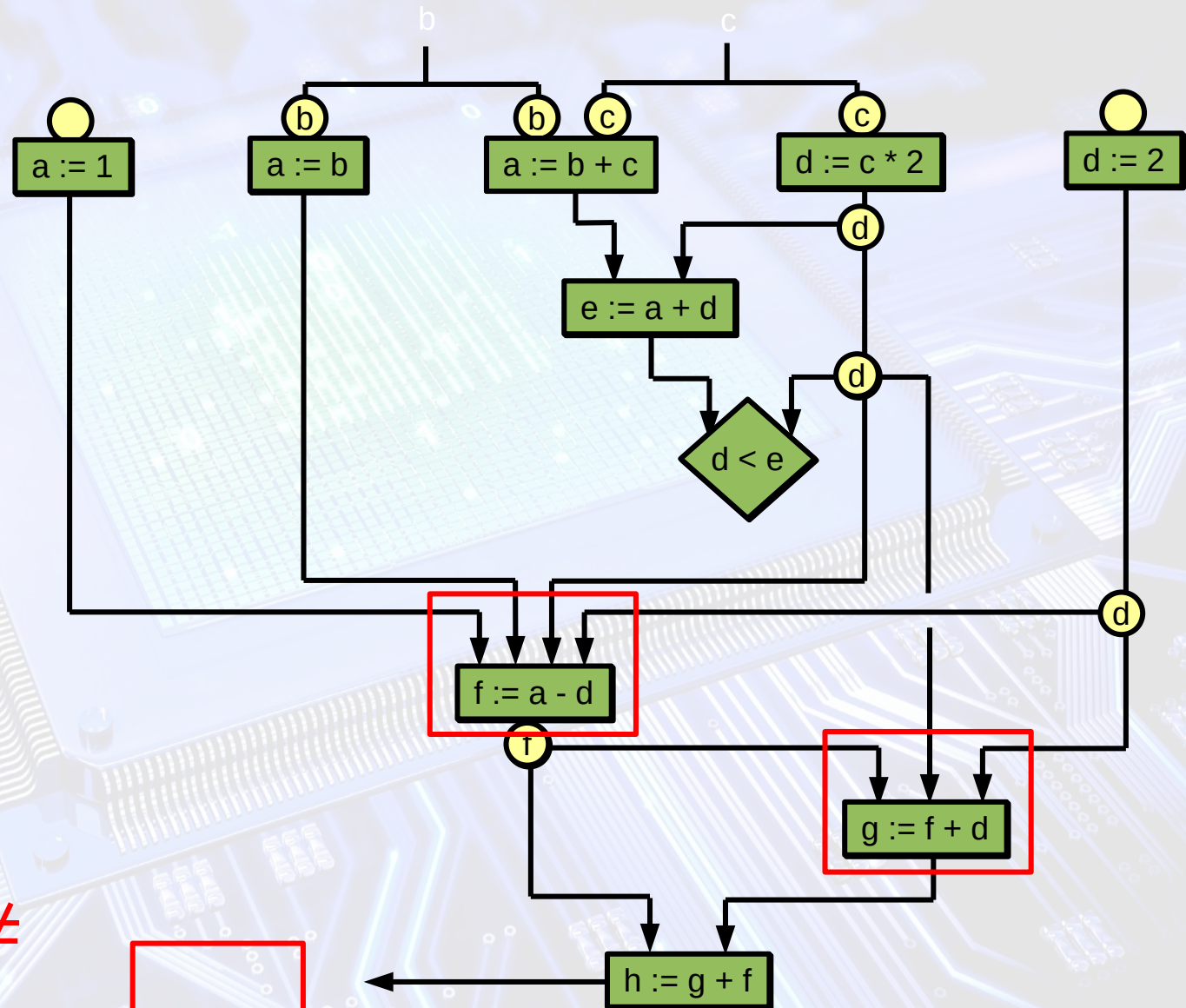
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  g := f + d  
  h := g + f  
}
```

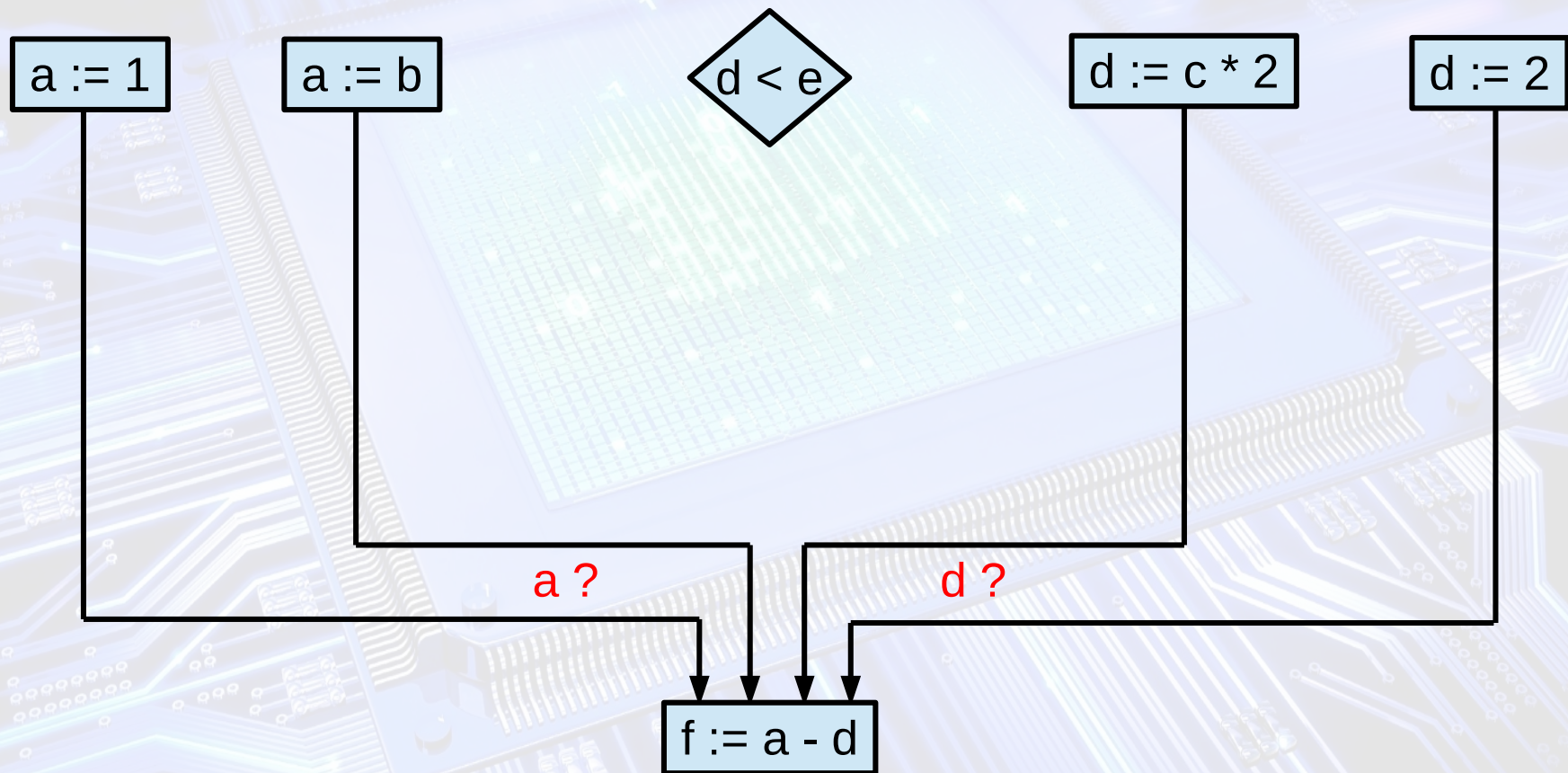
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Cause of the problem

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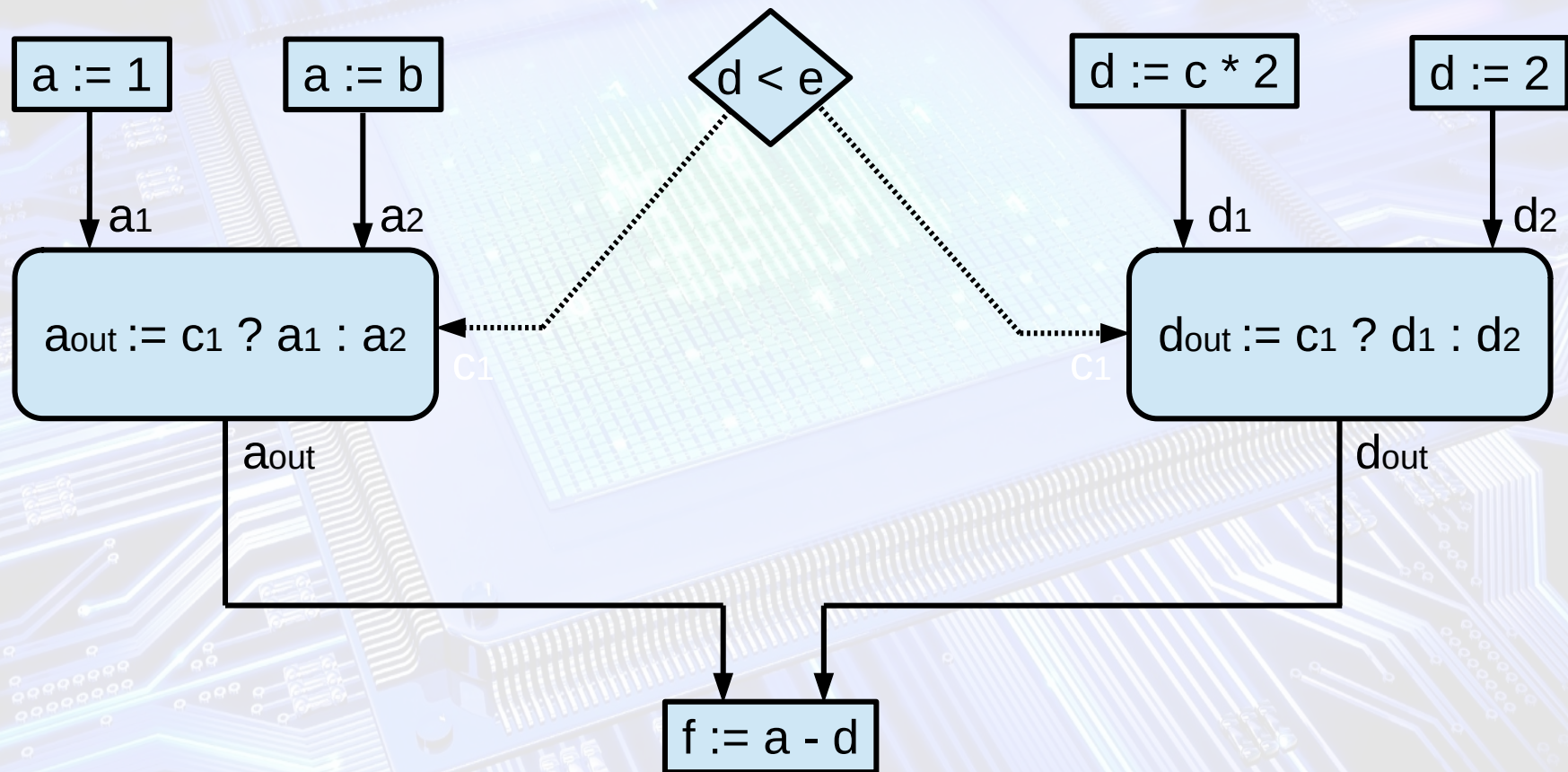
Data dependencies graph
The ambiguous entries
Insertion of the selectors



Solution to the problem

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Corrected data flow graph

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Data dependencies graph
The ambiguous entries
Insertion of the selectors

function (b, c)

{

$a := b + c$

$d := c * 2$

$e := a + d$

 if ($d < e$)

$a := 1$

 else

$a := b$

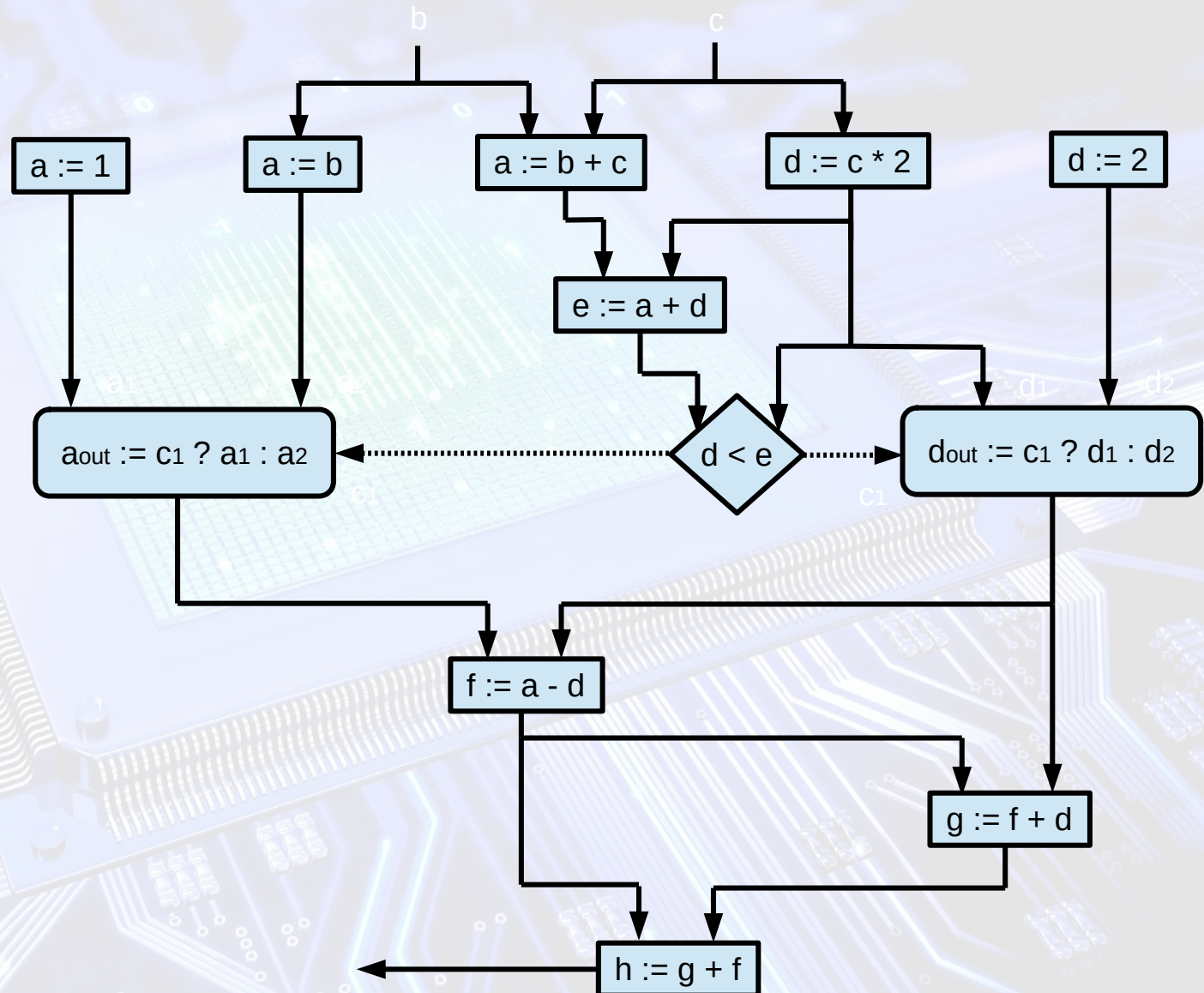
$d := 2$

$f := a - d$

$g := f + d$

$h := g + f$

}



Semantics are conserved

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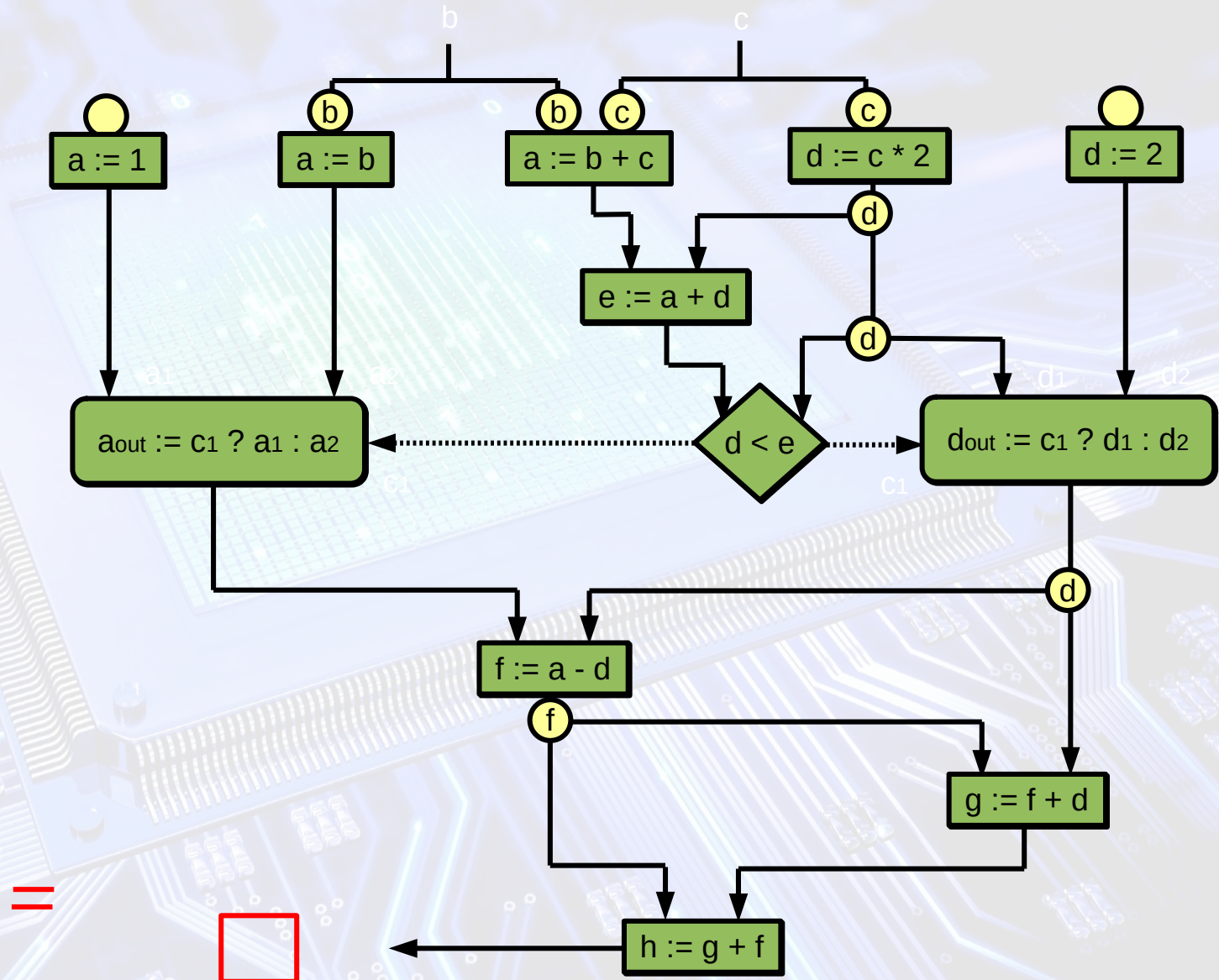
$d := 2$

$f := a - d$

$g := f + d$

$h := g + f$

}



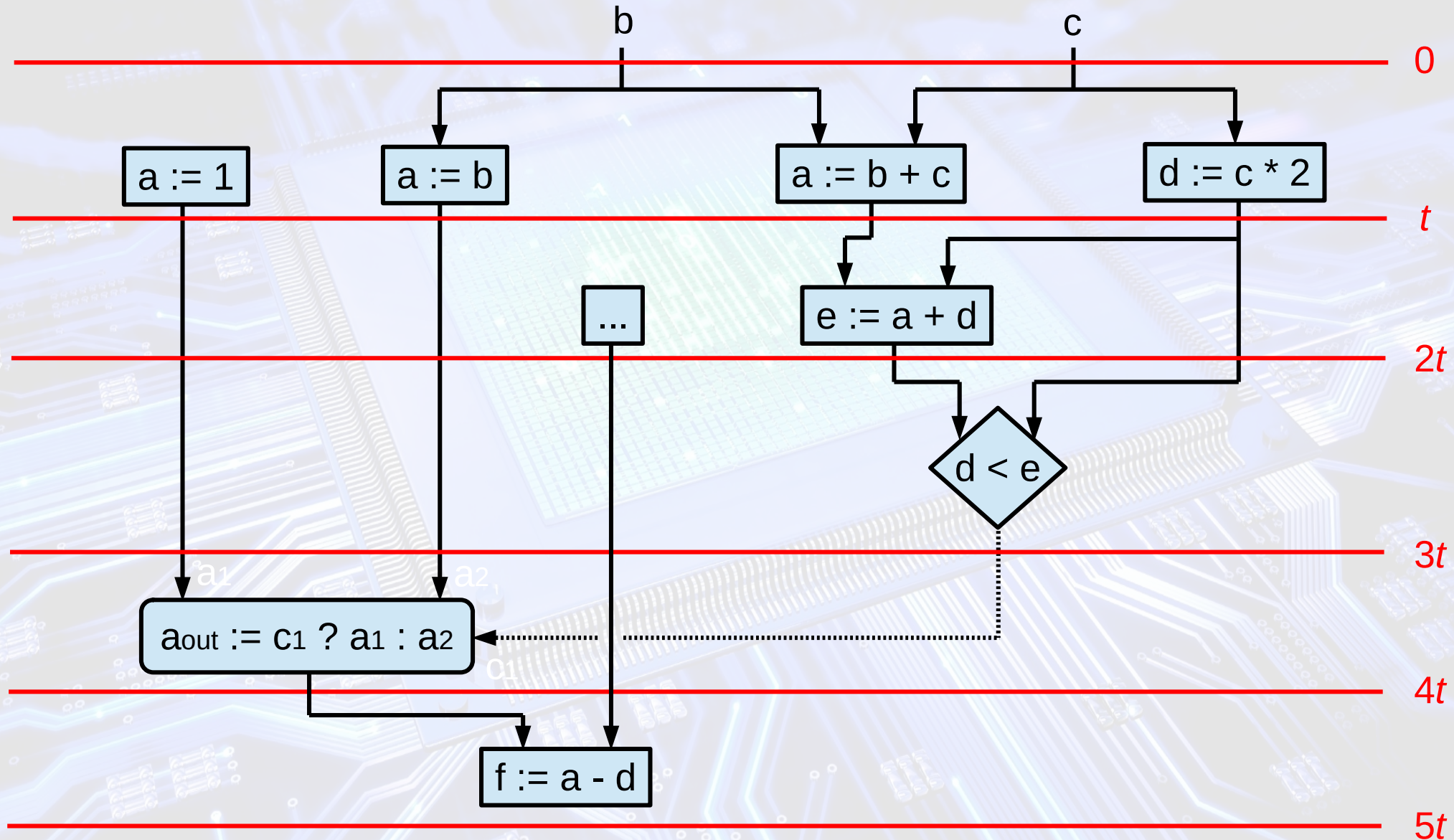
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Not enough condition information

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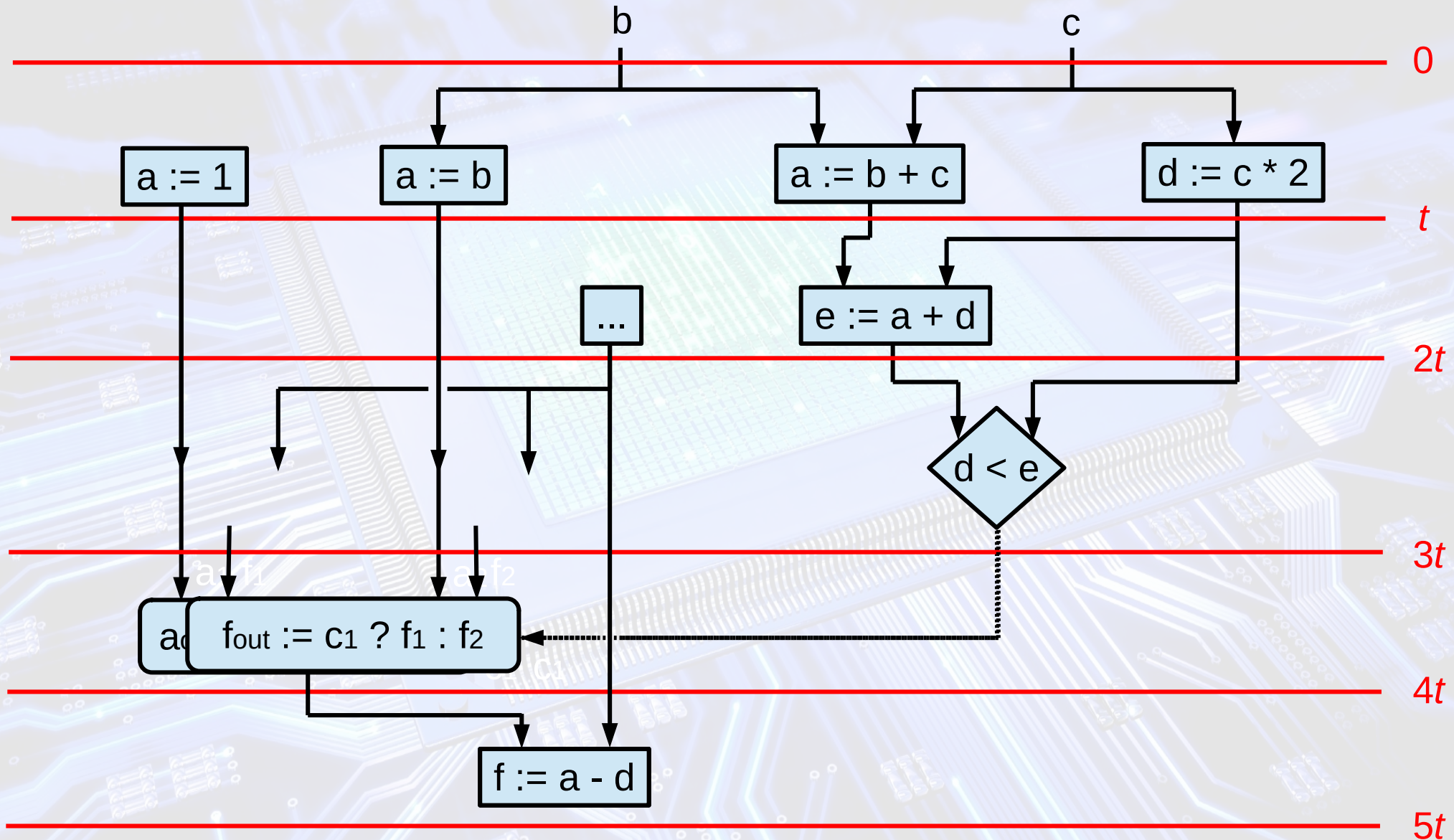
Time loss analysis
Reduction of the time loss



Duplication of the code

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Transformed data flow graph

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Time loss analysis
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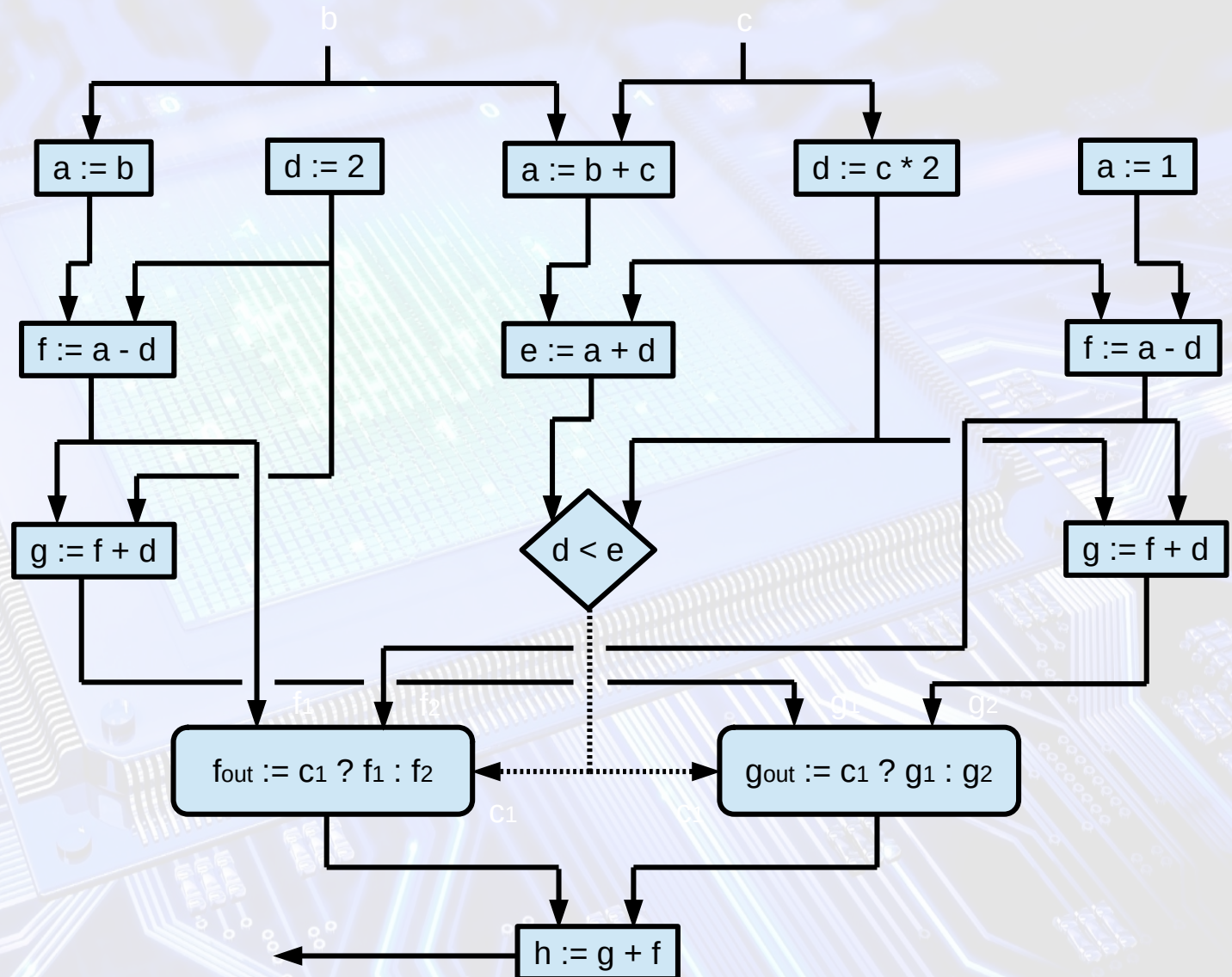
 d := 2

 f := a - d

 g := f + d

 h := g + f

}



Acceleration of the program

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Time loss analysis
Reduction of the time loss

function (b, c)

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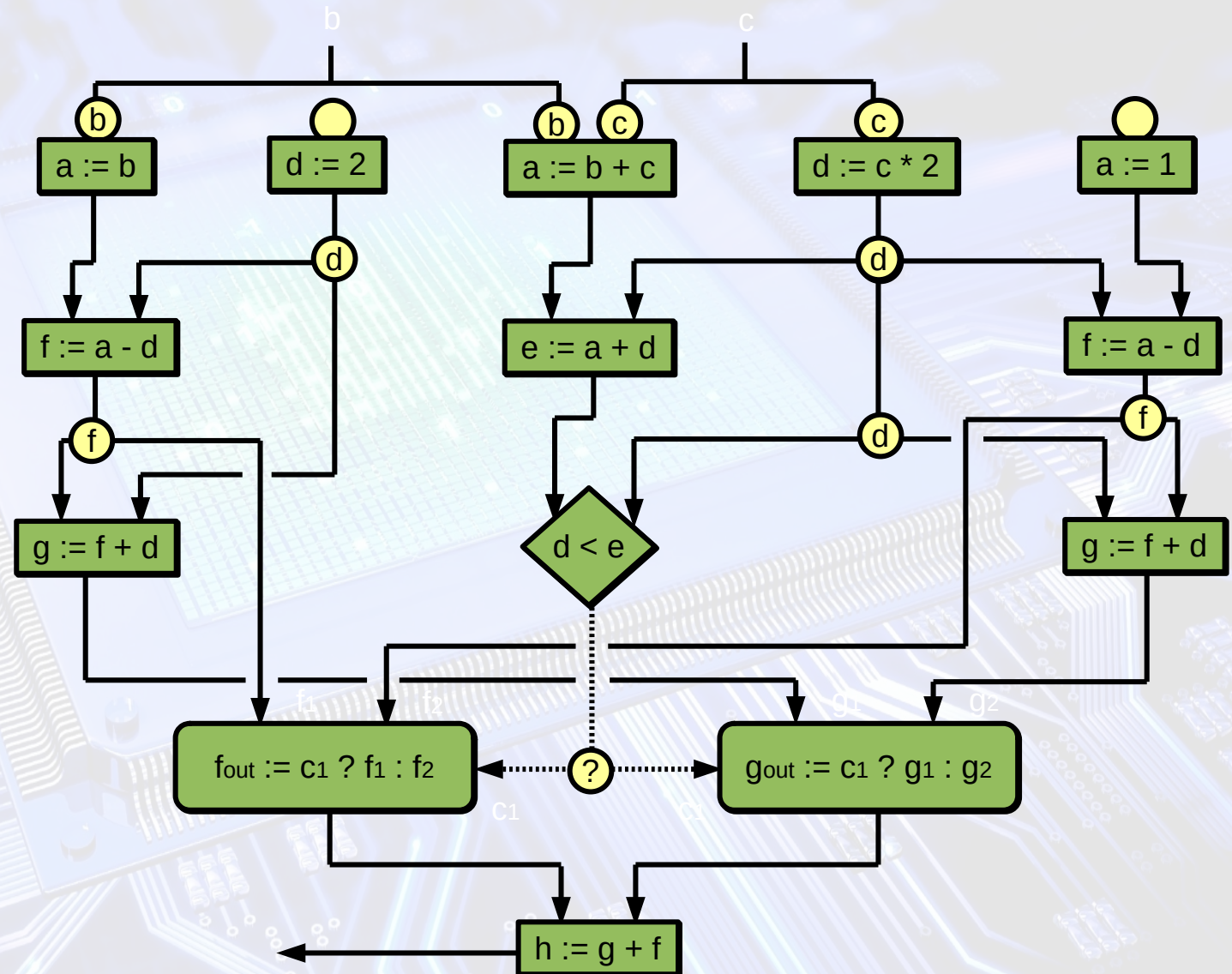
$d := 2$

$f := a - d$

$g := f + d$

$h := g + f$

}



My internship

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Summation
Future work
Questions

- Theoretical model
 - Real-time aware compilation technique
 - Multicore platforms
 - Strong assumptions
- ▶▶▶ More details in the internship report

Relaxation of the assumptions

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Questions

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Thank you

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Future work
Questions

